

SPCE 5065

HOMEWORK #3

DUE: Friday 3 July 2026

All homework is due in Canvas by 11:59:59 pm MT (Colorado Time) on the due date indicated on the Syllabus schedule. No late homework will be accepted.

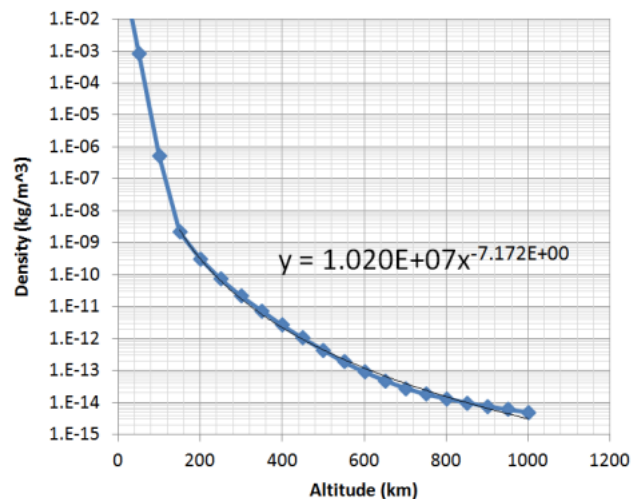
Students are allowed PARTIAL COLLABORATION on homework assignments. You are allowed to discuss qualitatively with other students the concepts in this course. However, copying or in any way using the written work of another person as well as relaying or receiving solutions via any means is strictly prohibited. The intent of this policy is to allow you to share ideas, discuss concepts, and clarify processes when needed. However, you must independently prepare the detailed solutions to homework problems and the Final Project.

Use course readings as a resource, but research additional sources as necessary to answer the questions. Please annotate your research with [1], [2], etc. and a sources page that describe where the information is located in AIAA format. Please email me with any questions or concerns.

Clearly state any assumptions you make.

1. A 100 kg satellite with an effective cross-sectional area of 1 m² is in a circular earth orbit at an altitude of 400 km. Its drag coefficient is 2.2. A rough estimate of the atmospheric density in the thermosphere is given by:

**High Atmosphere Density with
Trendline for >150 km**



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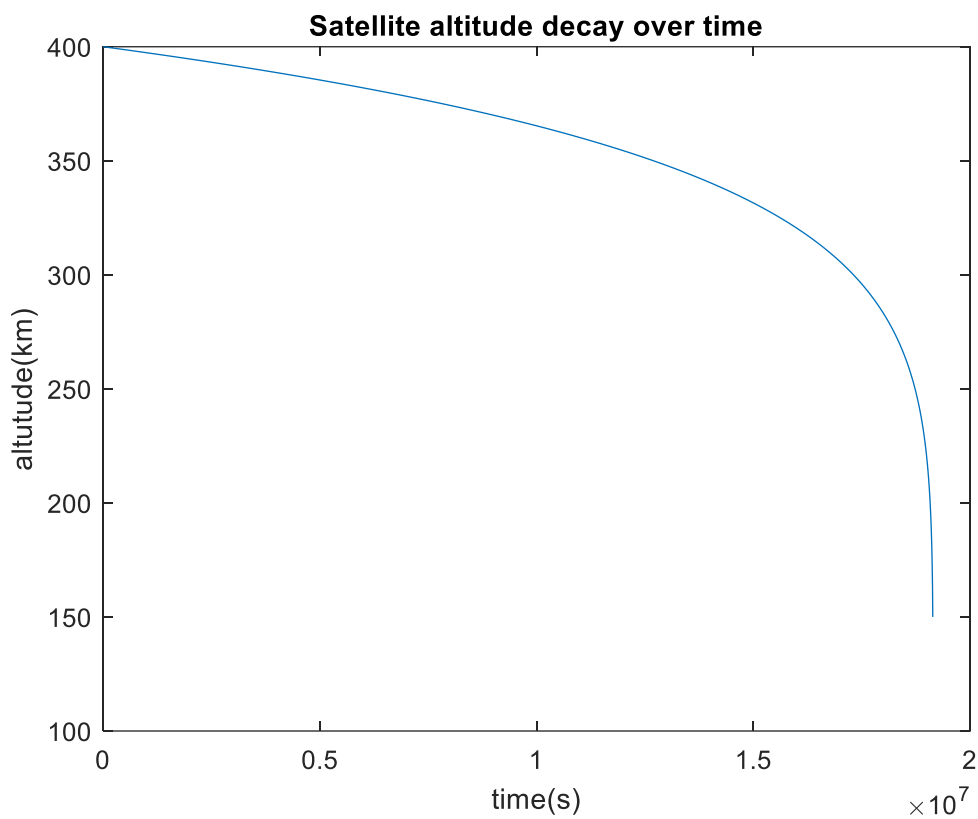
Estimate the satellite's lifetime if you assume the spacecraft deorbits at an altitude of 150 km. Do not assume an average R value as we did for the in-class exercise.

Solution:

Using material from lesson 5 in-class exercise:

$$\frac{da}{dt} = -\rho \frac{C_D A}{m} \sqrt{\mu a}$$

See attached Matlab Code



Lifetime estimate = 221.5 days

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2. How much drag makeup fuel is needed for the spacecraft in problem 1 to maintain its original orbit for one year? Assume average solar cycle conditions and monopropellant fuel with an $I_{sp} = 200$ seconds. Use your model from part 1 to find the atmospheric density.

$$F_{DRAG} = 0.5 * \rho * C_D * A * V^2$$

$$\rho = 2.2432 * 10^{-12} \text{ kg/m}^3$$

$$F_{DRAG} = 0.5 * 2.2432 * 10^{-12} \text{ kg/m}^3 * 2.2 * 1\text{m}^2 * \frac{398600.5 \frac{\text{km}^3}{\text{s}^2}}{6778 \text{ km}} * \frac{(1000 \text{ m})^2}{\text{km}^2}$$

$$F_{DRAG} = 1.45 \times 10^{-4} \text{ N}$$

$$m' = \frac{F \Delta t}{v'} = \frac{F \Delta t}{I_{sp} g_0} = \frac{1.45 \times 10^{-4} \text{ N} * 365 * 24 * 3600 \text{ seconds}}{200 \text{ seconds} * 9.81 \frac{\text{m}}{\text{s}^2}}$$

$$m' = 2.33 \text{ kg}$$

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3. Determine the erosion depth per year of a Kapton panel of a spacecraft oriented in the RAM direction (the side that points in the direction of the satellite's velocity vector) at an altitude of 450 km during low, medium, and high solar activity. The atomic-oxygen number densities are 6×10^6 , 2×10^7 , and 1×10^8 atoms/cm³, respectively.

From equation 3.40 page 85

$$\frac{dx}{dt} = (RE)\phi$$

Where:

For Kapton H, $RE = 3.1 \times 10^{-24}$ cm³/atom

$$\phi = \frac{AO \text{ atom}}{cm^2} \text{ (orbital velocity)}$$

For Low solar activity, per year the erosion depth will be

$$\frac{dx}{dt} = 3.1 \times 10^{-24} \frac{cm^3}{atom} * 6 \times 10^6 \frac{atoms}{cm^2} \sqrt{\frac{398600.5 \frac{km^3}{s^2}}{(6378 + 400)km}} * \frac{1000 * 100 \text{ cm}}{km} * \frac{365 * 24 * 3600 \text{ s}}{year}$$

Erosion depth per year = 4.5×10^{-4} cm

For Medium solar activity:

Erosion depth per year = 0.0015 cm

For High solar activity

Erosion depth per year = 0.0075 cm

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4. The Apollo command module has a volume of 5.9 m^3 and its atmosphere is 100% oxygen at a pressure of 5 psia and temperature of 21°C .

a. Determine the mass of oxygen present.

$$p = 5 \text{ psia} \times (1.01325 \times 10^5 \text{ N/m}^2)/(14.7 \text{ psia}) = 3.446 \times 10^4 \text{ N/m}^2$$

$$R = 8314.472 \text{ J/(kmol-K)}$$

$$T = 21 + 273.15 = 294.14 \text{ K}$$

$$V = 5.9 \text{ m}^3$$

$$pV = nRT$$

$$n = PV/RT = (3.446 \times 10^4 \text{ N/m}^2)(5.9 \text{ m}^3)/((8314.472 \text{ J/(kmol-K)})(294.15 \text{ K}))$$

$$n = 0.0831 \text{ kmol}$$

Molar mass M of molecular oxygen is

$$M = 2 \times 15.9995 = 31.9988 \text{ kg/kmol}$$

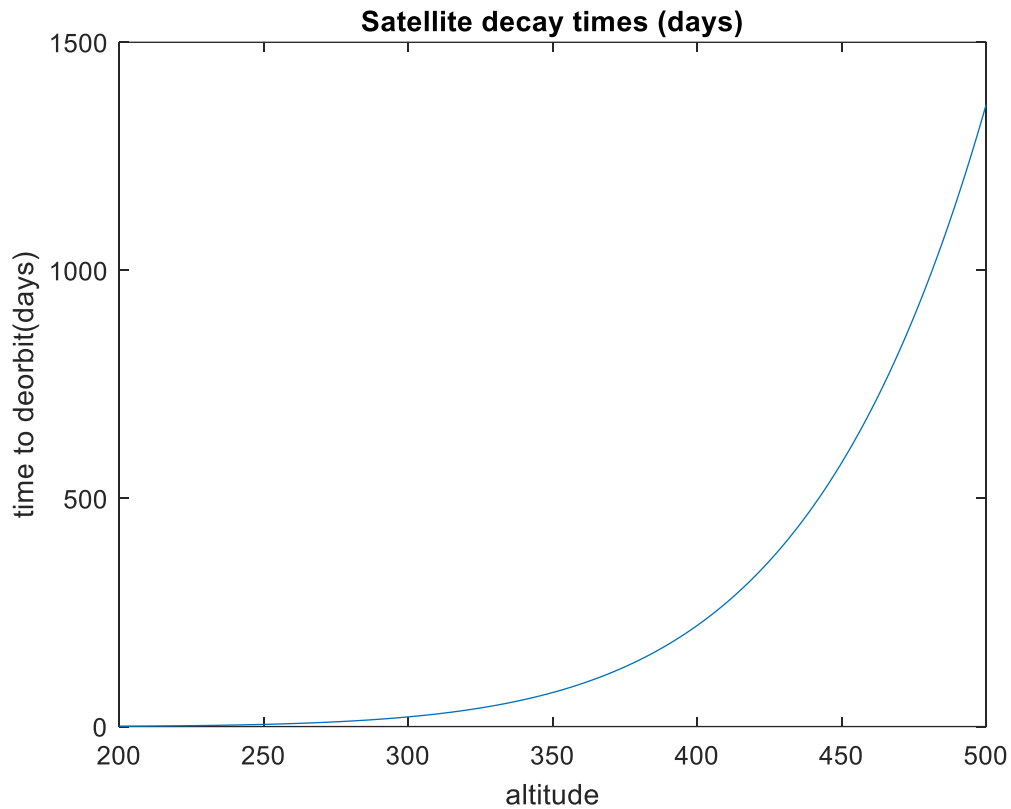
$$M = nM = 0.0831 \text{ kmol} \times 31.9988 \text{ kg/kmol} = 2.659 \text{ kg}$$

b. Would you recommend this atmosphere for a human vehicle bound for Mars? If not, what would you recommend?

No. Pure oxygen presents a fire hazard. I would recommend a mixture more like the earth's atmosphere of approximately 78% Nitrogen and 21% oxygen. We do not know what the long term (> 6 months) effects will be for breathing pure oxygen.

5. Create a plot of the lifetime of the satellite in problem 1 for all starting orbit altitudes between 200 and 600 km.

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6. The International Space Station uses Kapton for some applications.

a. What is Kapton used for on the ISS?

Answers may vary depending on the reference used. Common uses are for thermal control, atomic oxygen protection, and tape to repair electronic components.

b. Estimate the erosion of Kapton on the ISS for a mission of one year that is due to atomic oxygen.

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$$dx = RE\phi dt$$

$$RE = 10^{-24} \text{ cm}^3/\text{atom}$$

Average density = $1 \times 10^8 / \text{cm}^3$ (estimated from lesson 6 slide 29 for high solar activity)

$$V = \sqrt{\frac{\mu}{R}} = \sqrt{\frac{398600.5 \text{ km}^3/\text{s}^2}{6378 + 400 \text{ km}}} = 7.6686 \text{ km/s}$$

$$x = 10^{-24} \frac{\text{cm}^3}{\text{s}^2} \times 10^8 \frac{1}{\text{cm}^3} \times 7.6686 \frac{\text{km}}{\text{s}} \times \frac{1000 \text{ m}}{\text{km}} \times \frac{100 \text{ cm}}{\text{m}} \times 365 \text{ days} \times 24 \text{ hrs} \times 3600 \text{ s}$$

Erosion depth $\approx 2.418 \text{ } \mu\text{m}$

Answers may vary depending on assumptions.